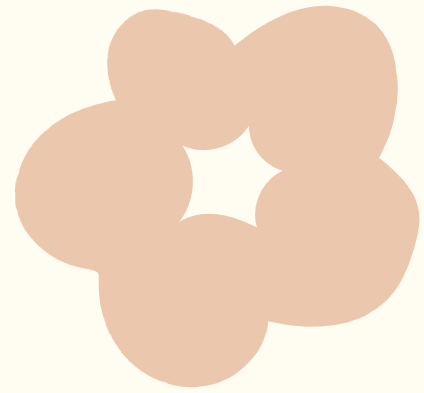
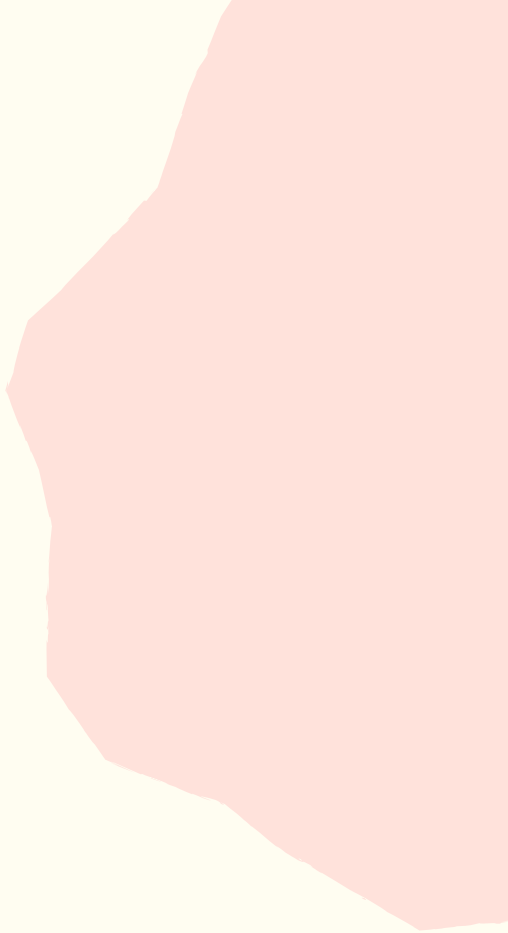
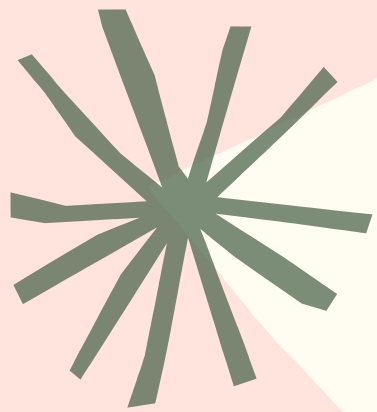




KAICHENG MA

Accompany Toy Bear

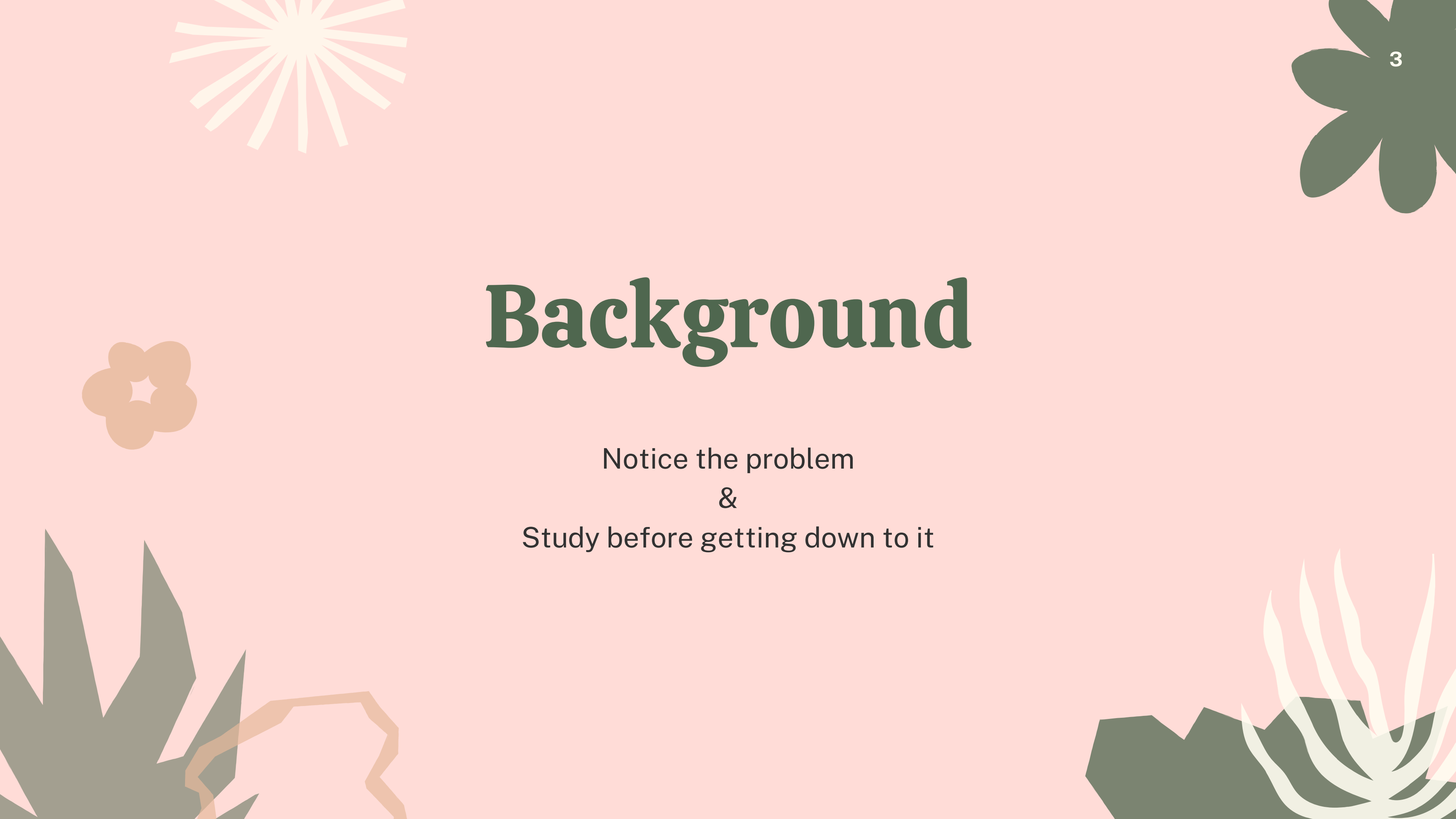
Adapting to the New Normal with Meditation



Topics for Discussion

The journey for the product

- **Background**
- **Product for Now**
- **What will the future be?**

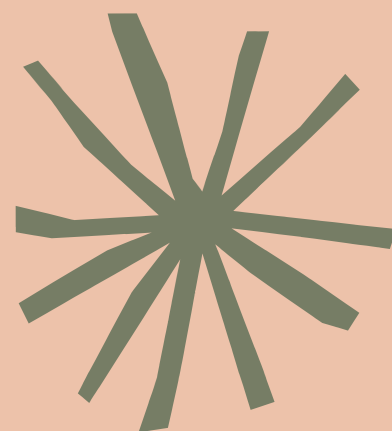


Background

Notice the problem
&
Study before getting down to it

Product Aim

Realize the problem
&
Set it as a goal



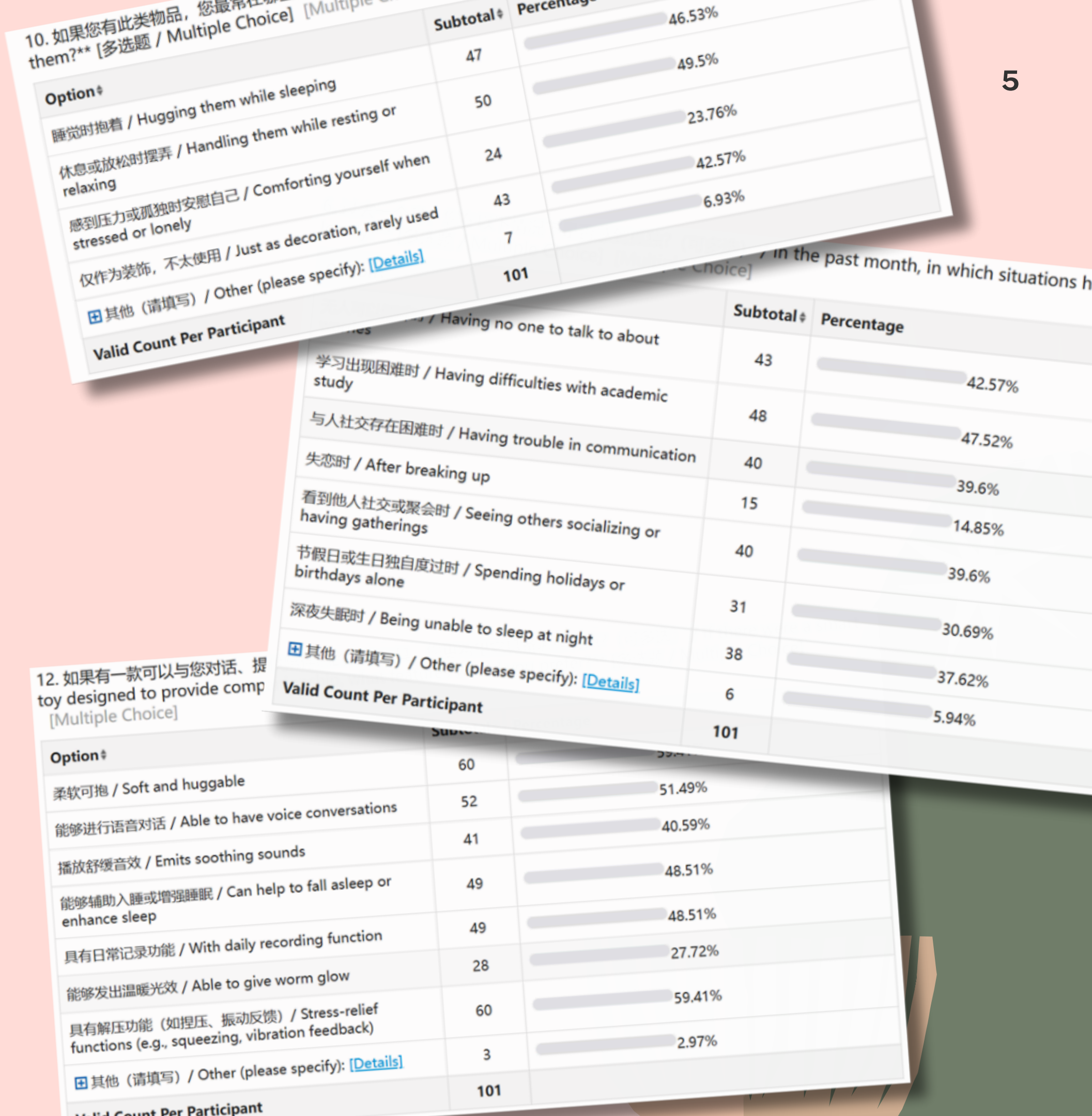
- 01 Ubiquitous:**
14%-24% of Students responded
“feel lonely often” / “very often”
- 02 Serious impacts:**
Affects a lot on mental health,
Sleep Quality, Suicidality
- 03 Worries (Goal):**
Worried about college students’
psychological health & life
happiness

Function setting

01 Sleep Aids:
Reduce the pressure & stress from the loneliness

02 AI voice dialogue:
Let the AI be the one you talk and share with

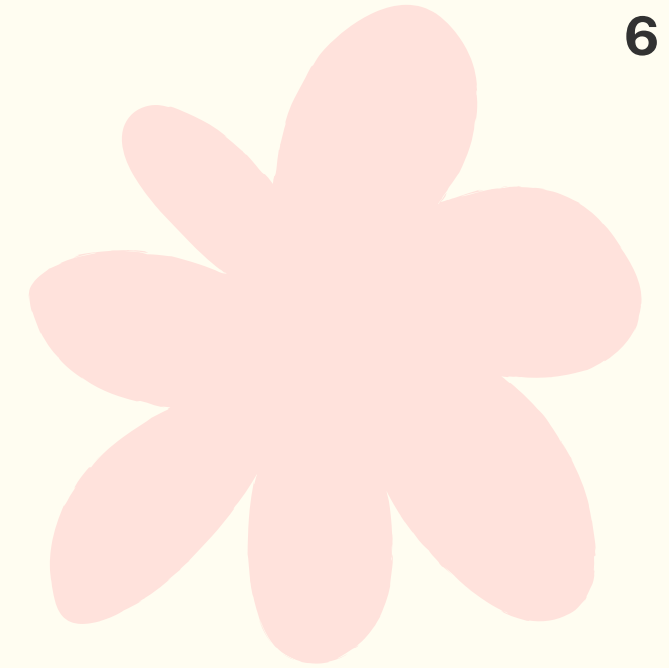
03 Companion and stress release:
An intimate appearance gives you a sense of companionship



Case Learning

6

Somnox

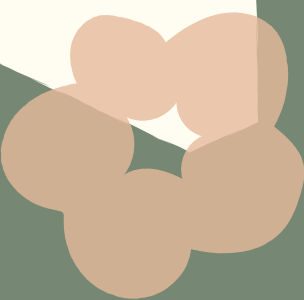
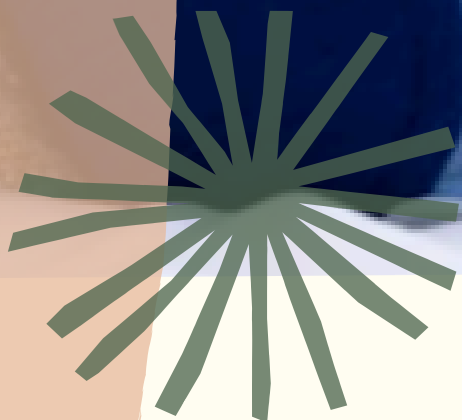
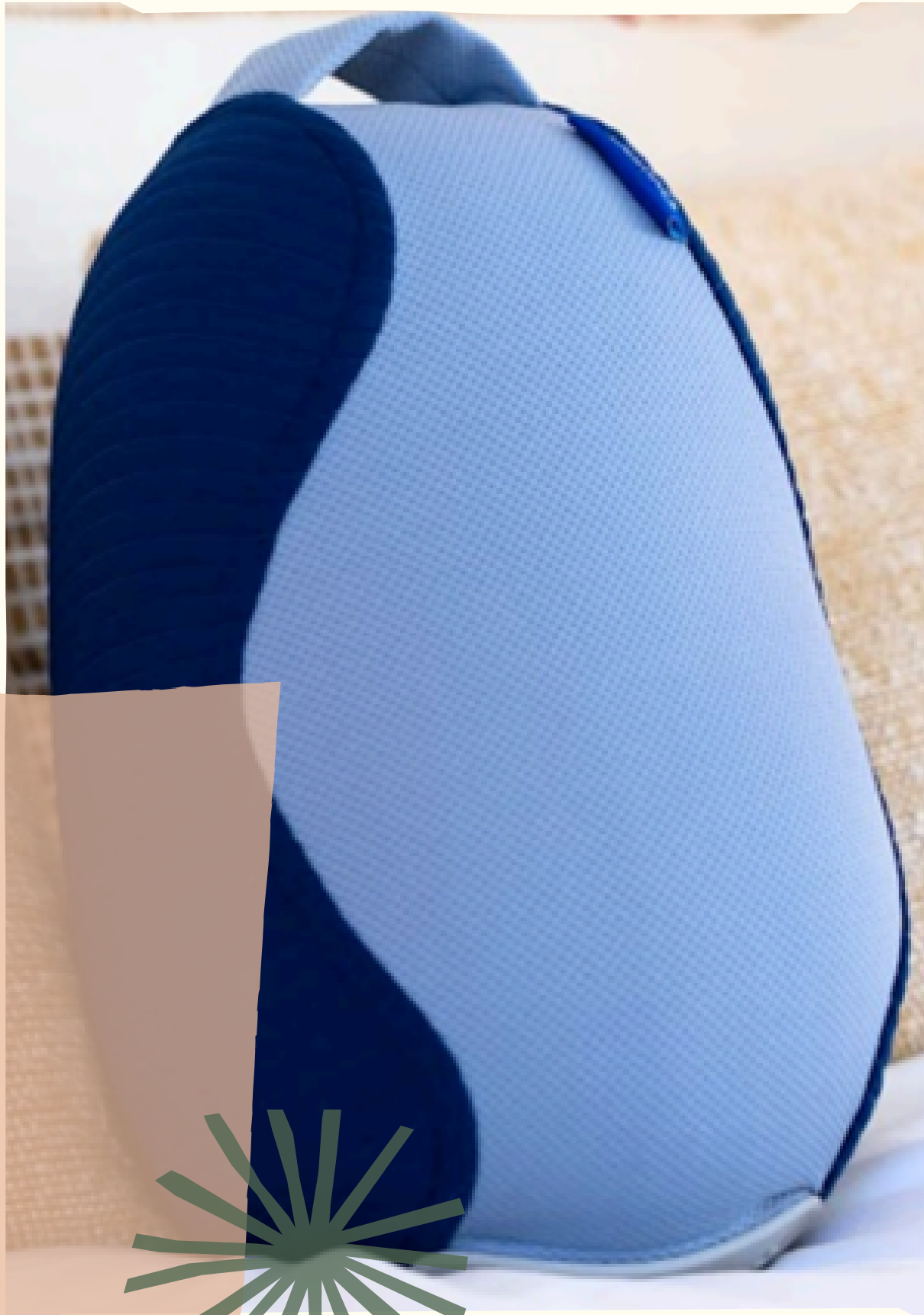


1.Slow-Pace Breathing Guidance:

-Enhance parasympathetic control (quicker, continuous)

2.For Our Product:

-Guidence → Vibrating motor



Case Learning

Moflin by Casio.

AI-powered behavioral patterns

Furry Appearance

Improved upon this: more intelligence, more interaction by adding interactional function with AI and specialized mechanism.



Case Learning

Teddy Bear and Stress Ball

- plush dolls can relieve stress to a certain extent
- bring people an emotional sense of security
- make people comfortable
- material to squeeze-release stress

To the
product

Material-appearance



Use

-Concrete Use Value:

(Provides psychological comfort,
addressing lack of communication and
insomnia)

Need

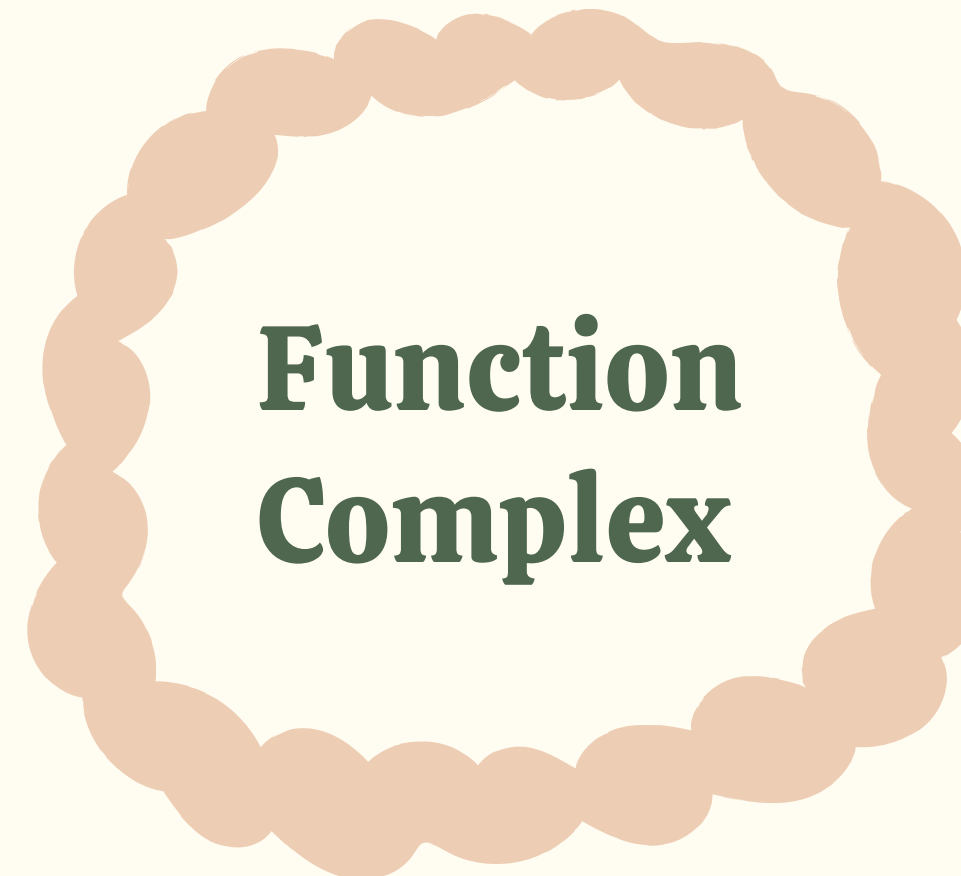
- Psychological Human Genuine Need:

(let lonely college students get
psychological help)

Telesis

-Strongly Fit in Society Background:

(Widespread phenomenon of loneliness)



Function Complex

Association

-Purposed Association Guidance:

(Lovely animal appearance gives
warm company)

Method

-Sufficient rationality:

- Tool: 3D modeling, Arduino
- Process: 3D Printing, AI Model
- Material: TPU etc.

Aesthetics

-move & please people, be meaningful:

(animal shapes and soft materials)

Product for NOW

MVP Design
&
Production process



Technological Implementations 1

modeling and realization



11

01

**Determine the main
appearance and
location of function**



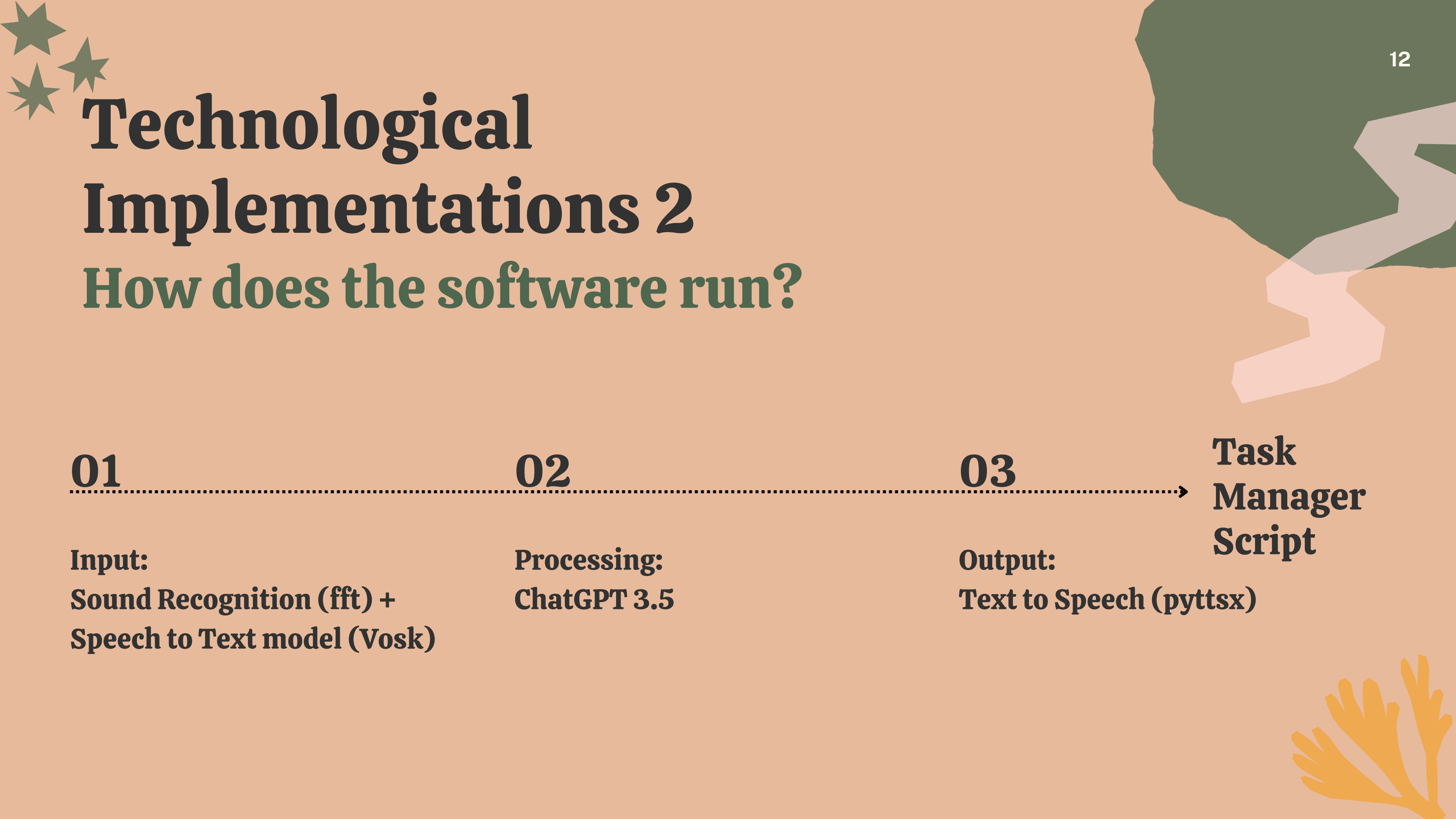
- make sure-cute/ comfortable
- position of the doll
- add a screen
- function position

02

process of modeling



- size
- scale
- suitable for printing
- Adjust the whole look



Technological Implementations 2

How does the software run?

01

Input:
**Sound Recognition (fft) +
Speech to Text model (Vosk)**

02

Processing:
ChatGPT 3.5

03

Output:
Text to Speech (pyttsx)

**Task
Manager
Script**

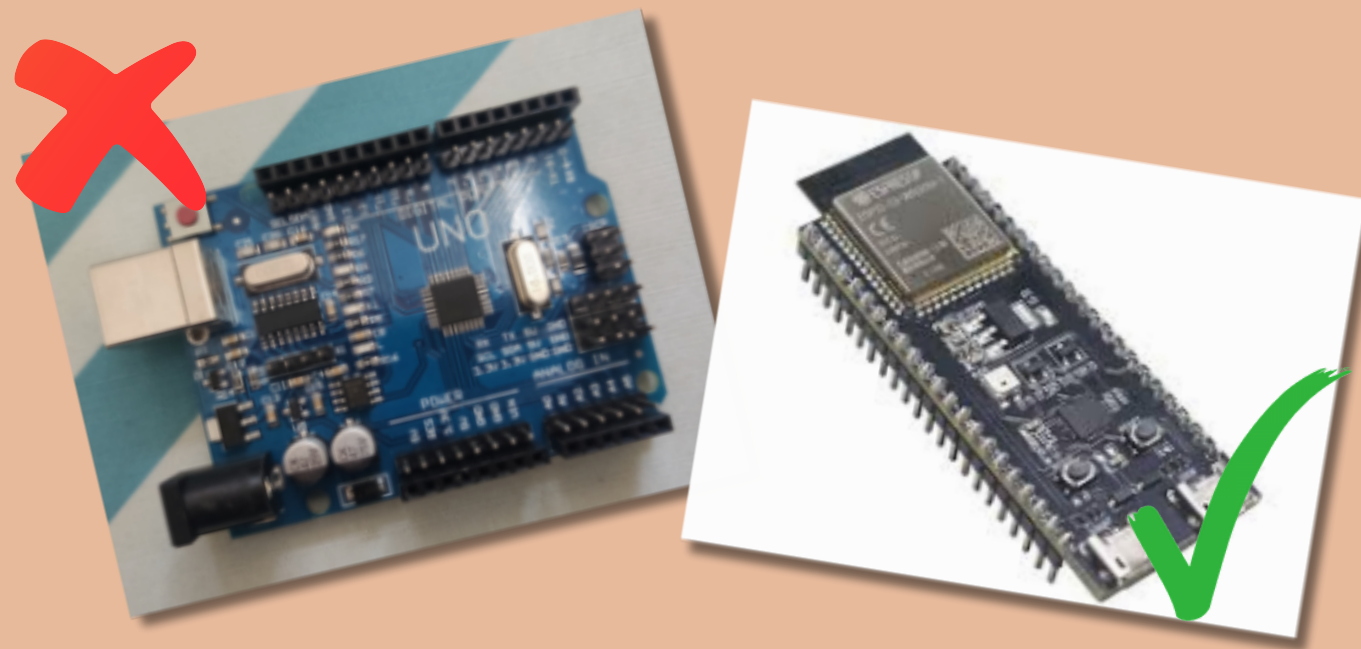
Technological Implementations 3

Microcontroller Setting

01

Microcontroller choice

Due to the necessity of I2S signal input and integration of AI:
Arduino UNO X ESP-32 S3 ✓



02

Programming environment construction

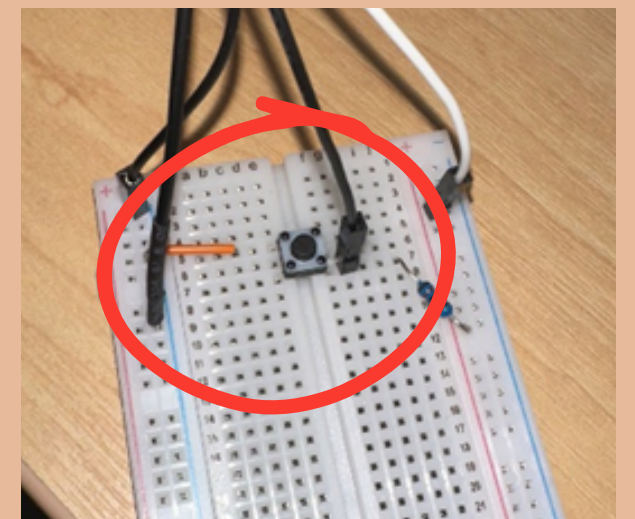
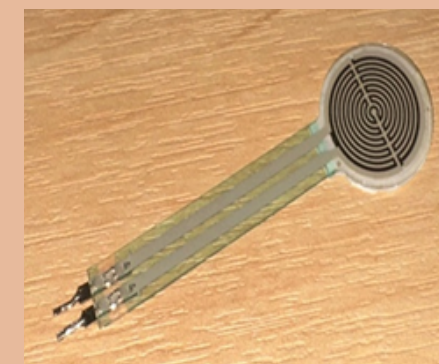
Arduino IDE ✓



03

Lack of pressure sensor

Risk of damage from improper voltage + unsold
Solution: **sensor** → **button**



Technological Implementations 3

Microcontroller Programming

01

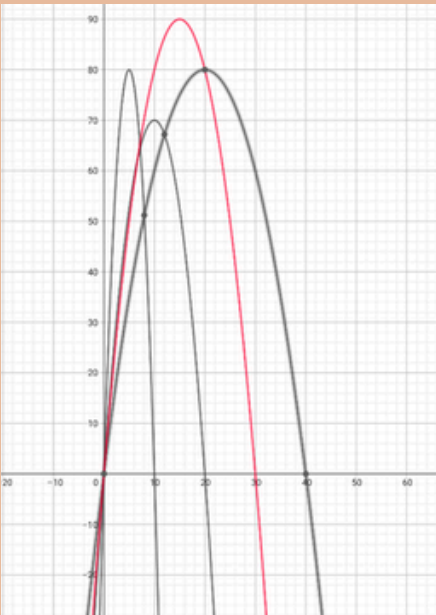
Vibrating Motor

$$f(x) = -\frac{16}{5}(x-5)^2 + 80 \quad \dots$$

$$g(x) = -\frac{1}{5}(x-20)^2 + 80 \quad \dots$$

$$h(x) = -\frac{7}{10}(x-10)^2 + 70 \quad \dots$$

$$p(x) = -\frac{2}{5}(x-15)^2 + 90 \quad \dots$$



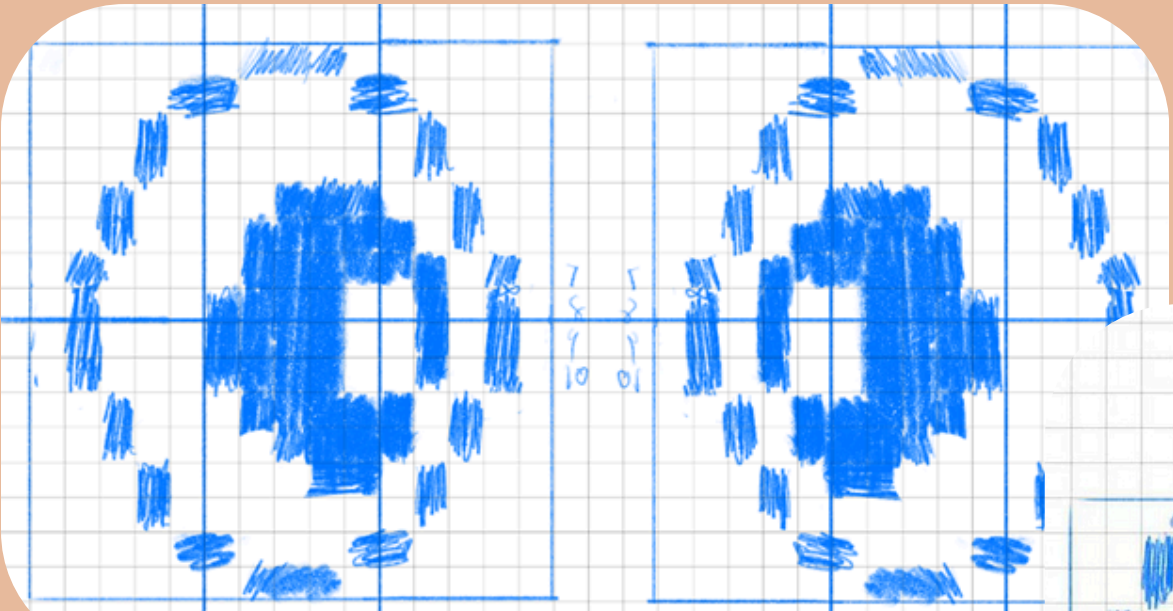
Test out the most comfortable vibration frequency for breathing guidance.



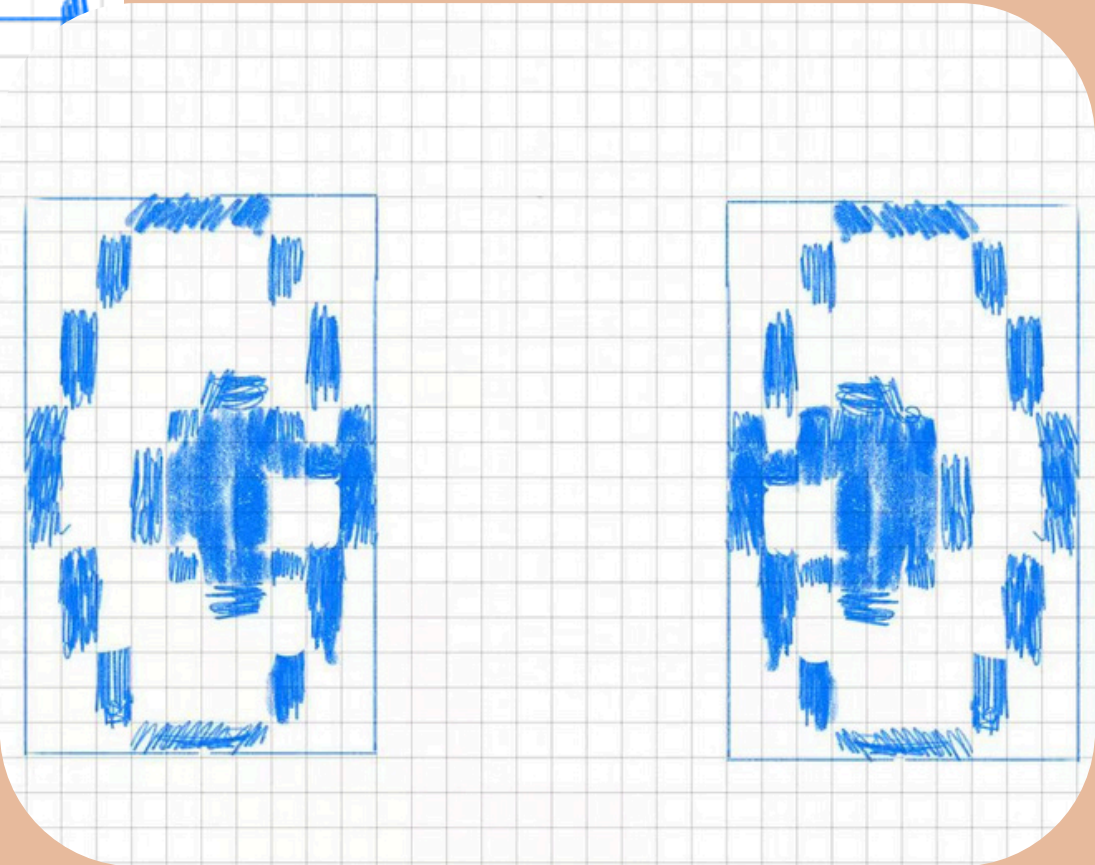
static → animation

```
//left normal eyes
uint8_t L1[8] = {0x00, 0x01, 0x02, 0x02, 0x04, 0x04, 0x08, 0x08};
uint8_t L2[8] = {0x0E, 0x11, 0x00, 0x00, 0x07, 0x0F, 0x0F, 0x1E};
uint8_t L3[8] = {0x00, 0x10, 0x08, 0x08, 0x04, 0x14, 0x1A, 0x0A};
uint8_t L4[8] = {0x08, 0x08, 0x04, 0x04, 0x02, 0x02, 0x01, 0x00};
uint8_t L5[8] = {0x1E, 0x1E, 0x0F, 0x07, 0x03, 0x00, 0x11, 0x0E};
uint8_t L6[8] = {0x0A, 0x0A, 0x14, 0x14, 0x08, 0x08, 0x10, 0x00};
```

Using hexadecimal number

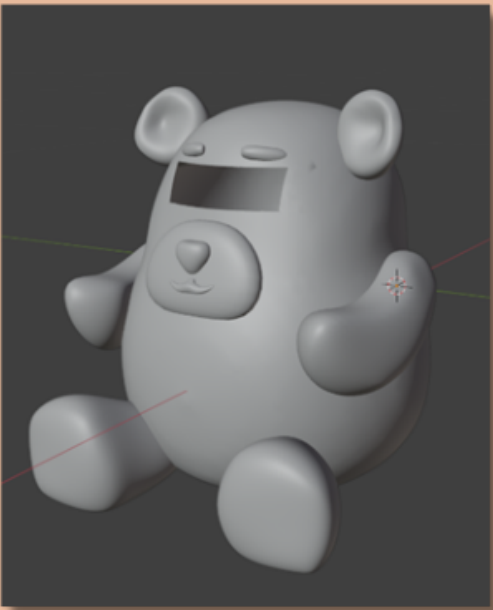


Eye Design 02

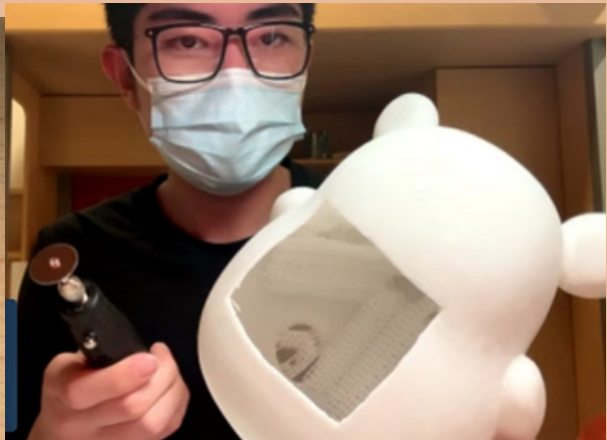
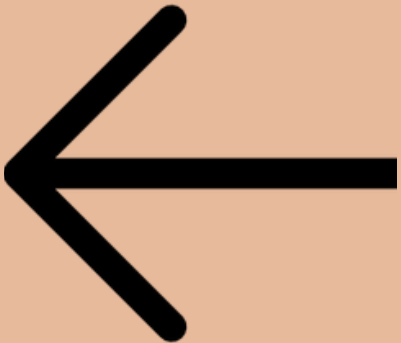


Technological Implementations 4

Assembling



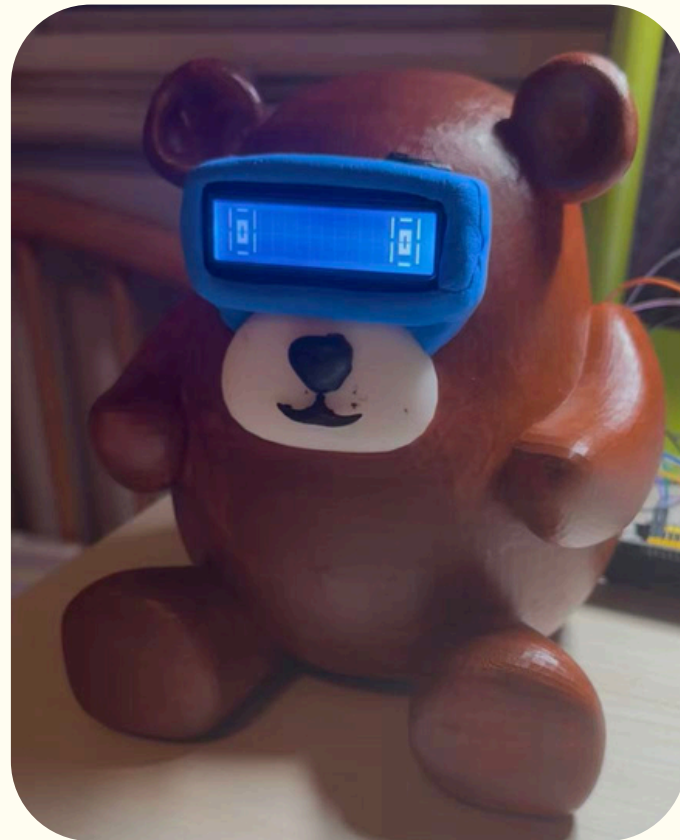
switch
design



MVP

16

Eye States



Breath Guidance with Light



Inhale



Exhale